```json
{
"name": "WF2_Booking_Orchestrator",
"nodes": [
{
"parameters": {},
"name": "Manual Trigger",
"type": "n8n-nodes-base.manualTrigger",
"typeVersion": 1,
"position": [0, 0],
"id": "manual_trigger"
},
{
"parameters": {
"path": "booking-orchestrator",
"httpMethod": "POST",
"responseMode": "lastNode",
"options": {}
},
"name": "Webhook",
"type": "n8n-nodes-base.webhook",
"typeVersion": 2.1,
"position": [0, 200],
"webhookId": "booking-orchestrator",
"id": "webhook_trigger"
},
{
"parameters": {
"jsCode": "// Generate Idempotency Key - FASE 1.2\n// Hash determinístico basado en datos de negocio (NO timestamp)\nconst input = $input.first().json;\n\n// Validar campos requeridos\nconst required = ['provider_id', 'service_id', 'start_time'];\nfor (const field of required) {\n  if (!input[field]) {\n    throw new Error(`Missing required field: ${field}`);\n  }\n}\n\n// Generar idempotency_key único para este booking\nconst idempotencyKey = `booking_${input.provider_id}_${input.service_id}_${input.start_time}_${input.customer_id || input.chat_id || 'anonymous'}`;\n\n// Sanitizar (SEC05)\nconst sanitizedKey = idempotencyKey.replace(/[^a-zA-Z0-9_:-]/g, '_').substring(0, 255);\n\n// PRESERVAR TODOS LOS DATOS ORIGINALES PARA REFERENCIA FUTURA\nreturn {\n  json: {\n    ...input,\n  idempotency_key: sanitizedKey,\n    _idempotency: { key: sanitizedKey, created_at: new Date().toISOString() }\n  }\n};"
},
"name": "Generate Idempotency Key",
"type": "n8n-nodes-base.code",
"typeVersion": 2,
"position": [250, 100],
"id": "generate_idempotency"
},
{
"parameters": {
"operation": "executeQuery",
"query": "SELECT booking_id, status, gcal_event_id FROM bookings WHERE idempotency_key = $1::text LIMIT 1;",
"options": {
"queryReplacement": "={{ $('Generate Idempotency Key').first().json.idempotency_key }}"
}
},
"name": "Check Idempotency DB",
"type": "n8n-nodes-base.postgres",
```

```
"typeVersion": 2.6,
"position": [500, 100],
"credentials": {
"postgres": { "id": "SFNQsmuu4zirZAnP", "name": "Postgres account" }
},
"id": "check_idempotency_db"
},
{
"parameters": {
"conditions": {
"string": [
{ "value": "", "operation": "notEmpty", "property": "={{ $json.booking_id }}" }
]
}
},
"name": "Is Duplicate?",
"type": "n8n-nodes-base.if",
"typeVersion": 2.3,
"position": [750, 100],
"id": "is_duplicate"
},
{
"parameters": {
"jsCode": "// Return Duplicate Booking\nconst existing = $('Check Idempotency
DB').first().json;\nconst original = $('Generate Idempotency Key').first().json;\n\nreturn
{\n  json: {\n    success: true,\n    error_code: null,\n    error_message: null,\n    data:
{\n      booking_id: existing.booking_id,\n      status: existing.status,\n
gcal_event_id: existing.gcal_event_id,\n      is_duplicate: true,\n      message: 'Booking
already exists (idempotent retry)'\n    },\n    _meta: { source: 'WF2_BOOKING_ORCHESTRATOR',
timestamp: new Date().toISOString(), version: '2.0.0', idempotency: 'duplicate' }\n  }\n};"
},
"name": "Return Duplicate",
"type": "n8n-nodes-base.code",
"typeVersion": 2,
"position": [1000, 200],
"id": "return_duplicate"
},
{
"parameters": {
"url": "https://n8n.stax.ink/webhook/check-availability",
"method": "POST",
```

```
"sendBody": true,
"bodyParameters": {
"parameters": [
{ "name": "provider_id", "value": "={{ $('Generate Idempotency Key').first().json.provider_id
}}" },
{ "name": "service_id", "value": "={{ $('Generate Idempotency Key').first().json.service_id
}}" },
{ "name": "start_time", "value": "={{ $('Generate Idempotency Key').first().json.start_time
}}" }
]
},
"options": { "timeout": 30000, "retryOnFail": true, "maxRetries": 3, "retryInterval": 2000 }
},
"name": "Check Availability",
"type": "n8n-nodes-base.httpRequest",
"typeVersion": 4.4,
```

"position": [1000, 0],
"id": "check_availability"
},
{
"parameters": {
"conditions": {
"boolean": [
{ "value": true, "operation": "equal", "property": "={{ $json.success }}" }
]
}
},
"name": "Is Available?",
"type": "n8n-nodes-base.if",
"typeVersion": 2.3,
"position": [1250, 0],
"id": "is_available"
},
{
"parameters": {
"jsCode": "// Return Not Available\nreturn {\n json: {\n   success: false,\n   error_code: 'SLOT_NOT_AVAILABLE',\n   error_message: 'Requested time slot is not available',\n   data: null,\n   _meta: { source: 'WF2_BOOKING_ORCHESTRATOR', timestamp: new Date().toISOString(), version: '2.0.0' }\n }\n};"
},
"name": "Return Not Available",
"type": "n8n-nodes-base.code",
"typeVersion": 2,
"position": [1500, 100],
"id": "return_not_available"
},
{
"parameters": {
"url": "https://n8n.stax.ink/webhook/acquire-lock",
"method": "POST",
"sendBody": true,
"bodyParameters": {
"parameters": [
{ "name": "provider_id", "value": "={{ $('Generate Idempotency Key').first().json.provider_id }}" },
{ "name": "start_time", "value": "={{ $('Generate Idempotency Key').first().json.start_time }}" },
{ "name": "lock_duration_minutes", "value": "5" }
]
},
"options": { "timeout": 30000, "retryOnFail": true, "maxRetries": 3, "retryInterval": 2000 }
},
"name": "Acquire Lock",
"type": "n8n-nodes-base.httpRequest",
"typeVersion": 4.4,
"position": [1500, 0],
"id": "acquire_lock"
},
{
"parameters": {
"url": "https://n8n.stax.ink/webhook/circuit-breaker/check",
"method": "POST",
"sendBody": true,
"bodyParameters": {

```
"parameters": [
{ "name": "service_id", "value": "google_calendar" },
{ "name": "action", "value": "check" }
]
},
"options": { "timeout": 10000 }
},
"name": "Check Circuit Breaker",
"type": "n8n-nodes-base.httpRequest",
"typeVersion": 4.4,
"position": [1750, 0],
"id": "check_circuit_breaker"
},
{
```

```
"parameters": {
"conditions": {
"boolean": [
{ "value": true, "operation": "equal", "property": "={{ $json.data.allowed }}" }
]
}
},
"name": "Is GCal Available?",
"type": "n8n-nodes-base.if",
"typeVersion": 2.3,
"position": [2000, 0],
"id": "is_gcal_available"
},
{
"parameters": {
"operation": "create",
"calendar": {
"__rl": true,
"value": "dev.n8n.stax@gmail.com",
"mode": "list",
"cachedResultName": "dev.n8n.stax@gmail.com"
},
"event": {
"summary": "={{ $('Generate Idempotency Key').first().json.event_title || 'Booking
Appointment' }}",
"start": "={{ $('Generate Idempotency Key').first().json.start_time }}",
"end": "={{ $('Generate Idempotency Key').first().json.end_time || new Date(new
Date($('Generate Idempotency Key').first().json.start_time).getTime() +
3600000).toISOString() }}",
"description": "={{ JSON.stringify({ booking_id: $('Generate Idempotency
Key').first().json.idempotency_key, customer: $('Generate Idempotency
Key').first().json.customer_id || $('Generate Idempotency Key').first().json.chat_id }) }}"
},
"options": {}
},
"name": "Create GCal Event",
"type": "n8n-nodes-base.googleCalendar",
"typeVersion": 1.3,
"position": [2250, -100],
"credentials": {
"googleCalendarOAuth2Api": { "id": "NDk9hJZ9MTokyiAh", "name": "Google Calendar account" }
},
"id": "create_gcal_event"
```

```
},
{
"parameters": {
"url": "https://n8n.stax.ink/webhook/circuit-breaker/record",
"method": "POST",
"sendBody": true,
"bodyParameters": {
"parameters": [
{ "name": "service_id", "value": "google_calendar" },
{ "name": "success", "value": "true" }
]
},
"options": {}
},
"name": "Record GCal Success",
"type": "n8n-nodes-base.httpRequest",
"typeVersion": 4.4,
"position": [2500, -100],
"id": "record_gcal_success"
},
{
"parameters": {
"jsCode": "// Prepare DB Insert - Reference original data\nconst original = $('Generate
Idempotency Key').first().json;\nconst gcal = $('Create GCal Event').first().json;\n\nreturn
{\n  json: {\n    provider_id: original.provider_id,\n    service_id: original.service_id,\n
start_time: original.start_time,\n    idempotency_key: original.idempotency_key,\n
gcal_event_id: gcal.id || gcal.eventId,\n    customer_id: original.customer_id || '',\n
chat_id: original.chat_id || ''\n  }\n};"
},
"name": "Prepare DB Insert",
"type": "n8n-nodes-base.code",
"typeVersion": 2,
"position": [2750, -100],
"id": "prepare_db_insert"
},
{
"parameters": {
"operation": "executeQuery",
"query": "INSERT INTO bookings (provider_id, service_id, start_time, idempotency_key,
gcal_event_id, customer_id, chat_id, status, created_at) VALUES ($1::int, $2::int,
$3::timestamp, $4::text, $5::text, $6::text, $7::text, 'confirmed', NOW()) RETURNING
booking_id;",
"options": {
"queryReplacement": "={{ $('Prepare DB Insert').first().json.provider_id }}, {{ $('Prepare DB
Insert').first().json.service_id }}, {{ $('Prepare DB Insert').first().json.start_time }}, {{
$('Prepare DB Insert').first().json.idempotency_key }}, {{ $('Prepare DB
Insert').first().json.gcal_event_id }}, {{ $('Prepare DB Insert').first().json.customer_id
}}, {{ $('Prepare DB Insert').first().json.chat_id }}"
}
},
"name": "Create DB Booking",
"type": "n8n-nodes-base.postgres",
"typeVersion": 2.6,
"position": [3000, -100],
```

```
"credentials": {
"postgres": { "id": "SFNQsmuu4zirZAnP", "name": "Postgres account" }
},
```

"id": "create_db_booking"
},
{
"parameters": {
"url": "https://n8n.stax.ink/webhook/release-lock",
"method": "POST",
"sendBody": true,
"bodyParameters": {
"parameters": [
{ "name": "provider_id", "value": "={{ $('Generate Idempotency Key').first().json.provider_id
}}" },
{ "name": "start_time", "value": "={{ $('Generate Idempotency Key').first().json.start_time
}}" }
]
},
"options": {},
"continueOnFail": true
},
"name": "Release Lock",
"type": "n8n-nodes-base.httpRequest",
"typeVersion": 4.4,
"position": [3250, -100],
"id": "release_lock"
},
{
"parameters": {
"jsCode": "// Standard Contract Output - Booking Success\nconst db = $('Create DB
Booking').first().json;\nconst gcal = $('Create GCal Event').first().json;\nconst original =
$('Generate Idempotency Key').first().json;\n\nreturn {\n json: {\n   success: true,\n
error_code: null,\n   error_message: null,\n   data: {\n   booking_id: db.booking_id,\n
gcal_event_id: gcal.id || gcal.eventId,\n   idempotency_key: original.idempotency_key,\n
provider_id: original.provider_id,\n   service_id: original.service_id,\n   start_time:
original.start_time,\n   status: 'confirmed',\n   is_duplicate: false\n   },\n
_meta: { source: 'WF2_BOOKING_ORCHESTRATOR', timestamp: new Date().toISOString(), version:
'2.0.0', idempotency: 'new' }\n }\n};"
},
"name": "Standard Contract Output",
"type": "n8n-nodes-base.code",
"typeVersion": 2,
"position": [3500, -100],
"id": "standard_contract"
},
{
"parameters": {
"url": "https://n8n.stax.ink/webhook/circuit-breaker/record",
"method": "POST",
"sendBody": true,
"bodyParameters": {
"parameters": [
{ "name": "service_id", "value": "google_calendar" },
{ "name": "success", "value": "false" },
{ "name": "error_message", "value": "Circuit breaker OPEN" }
]
},
"options": {},
"continueOnFail": true
},
"name": "Record GCal Failure",

```
"type": "n8n-nodes-base.httpRequest",
"typeVersion": 4.4,
"position": [2250, 100],
"id": "record_gcal_failure"
},
{
"parameters": {
"url": "https://n8n.stax.ink/webhook/release-lock",
"method": "POST",
"sendBody": true,
"bodyParameters": {
"parameters": [
{ "name": "provider_id", "value": "={{ $('Generate Idempotency Key').first().json.provider_id
}}" },
{ "name": "start_time", "value": "={{ $('Generate Idempotency Key').first().json.start_time
}}" }
]
},
"options": {},
"continueOnFail": true
},
"name": "Release Lock on Error",
"type": "n8n-nodes-base.httpRequest",
"typeVersion": 4.4,
"position": [2500, 100],
"id": "release_lock_error"
},
{
"parameters": {
"url": "https://n8n.stax.ink/webhook/dlq/add",
"method": "POST",
```

```json
"sendBody": true,
"bodyParameters": {
"parameters": [
{ "name": "failure_reason", "value": "gcal_circuit_breaker_open" },
{ "name": "error_message", "value": "Google Calendar unavailable (circuit breaker open)" },
{ "name": "original_payload", "value": "={{ JSON.stringify($('Generate Idempotency
Key').first().json) }}" },
{ "name": "context_data", "value": "={{ JSON.stringify({ workflow_id:
'WF2_Booking_Orchestrator', execution_id: $execution.id, failed_at: new Date().toISOString()
}) }}" },
{ "name": "provider_id", "value": "={{ $('Generate Idempotency Key').first().json.provider_id
}}" },
{ "name": "service_id", "value": "={{ $('Generate Idempotency Key').first().json.service_id
}}" },
{ "name": "start_time", "value": "={{ $('Generate Idempotency Key').first().json.start_time
}}" },
{ "name": "customer_id", "value": "={{ $('Generate Idempotency Key').first().json.customer_id
}}" },
{ "name": "idempotency_key", "value": "={{ $('Generate Idempotency
Key').first().json.idempotency_key }}" }
]
},
"options": {}
},
"name": "Send to DLQ",
"type": "n8n-nodes-base.httpRequest",
```

"typeVersion": 4.4,
"position": [2750, 100],
"id": "send_to_dlq"
},
{
"parameters": {
"jsCode": "// Return Circuit Breaker Error\nreturn {\n json: {\n   success: false,\n
error_code: 'CIRCUIT_BREAKER_OPEN',\n   error_message: 'Google Calendar temporarily
unavailable. Booking added to retry queue.',\n   data: { will_retry: true },\n   _meta: {
source: 'WF2_BOOKING_ORCHESTRATOR', timestamp: new Date().toISOString(), version: '2.0.0' }\n
}\n};"
},
"name": "Return CB Error",
"type": "n8n-nodes-base.code",
"typeVersion": 2,
"position": [3000, 100],
"id": "return_cb_error"
}
],
"connections": {
"Manual Trigger": { "main": [[{ "node": "Generate Idempotency Key", "type": "main", "index":
0 }]] },
"Webhook": { "main": [[{ "node": "Generate Idempotency Key", "type": "main", "index": 0 }]]
},
"Generate Idempotency Key": { "main": [[{ "node": "Check Idempotency DB", "type": "main",
"index": 0 }]] },
"Check Idempotency DB": { "main": [[{ "node": "Is Duplicate?", "type": "main", "index": 0 }]]
},
"Is Duplicate?": { "main": [[{ "node": "Return Duplicate", "type": "main", "index": 0 }], [{
"node": "Check Availability", "type": "main", "index": 0 }]] },
"Check Availability": { "main": [[{ "node": "Is Available?", "type": "main", "index": 0 }]]
},
"Is Available?": { "main": [[{ "node": "Acquire Lock", "type": "main", "index": 0 }], [{
"node": "Return Not Available", "type": "main", "index": 0 }]] },
"Acquire Lock": { "main": [[{ "node": "Check Circuit Breaker", "type": "main", "index": 0 }]]
},
"Check Circuit Breaker": { "main": [[{ "node": "Is GCal Available?", "type": "main", "index":
0 }]] },
"Is GCal Available?": { "main": [[{ "node": "Create GCal Event", "type": "main", "index": 0
}], [{ "node": "Record GCal Failure", "type": "main", "index": 0 }]] },
"Create GCal Event": { "main": [[{ "node": "Record GCal Success", "type": "main", "index": 0
}]] },
"Record GCal Success": { "main": [[{ "node": "Prepare DB Insert", "type": "main", "index": 0
}]] },
"Prepare DB Insert": { "main": [[{ "node": "Create DB Booking", "type": "main", "index": 0
}]] },
"Create DB Booking": { "main": [[{ "node": "Release Lock", "type": "main", "index": 0 }]] },
"Release Lock": { "main": [[{ "node": "Standard Contract Output", "type": "main", "index": 0
}]] },
"Record GCal Failure": { "main": [[{ "node": "Release Lock on Error", "type": "main",
"index": 0 }]] },
"Release Lock on Error": { "main": [[{ "node": "Send to DLQ", "type": "main", "index": 0 }]]
},
"Send to DLQ": { "main": [[{ "node": "Return CB Error", "type": "main", "index": 0 }]] }
},
"settings": { "executionOrder": "v1" },
"active": false,
"versionId": "wf2-v2.0.0-rewritten"

}